

An Exploratory Study on Learning in 3D Multi-user Virtual Environments

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ABSTRACT

In this study, we propose and evaluate a desktop 3D multi-user virtual learning system for the study of butterflies. Multiple users can use avatars to learn and communicate through voice. Results indicate that the system received positive ratings in terms of perception, system effectiveness, learning experience, enjoyment, and attractiveness. However, areas for improvement were identified. The findings have practical implications for the design of educational multi-user virtual reality.

Keywords: Educational Virtual Reality, Multi-user Virtual Reality, Desktop VR, MUVE, Evaluation

Index Terms: Human-centered computing—Virtual Reality;

1 INTRODUCTION

The integration of virtual reality (VR) technology into the field of education has been widely regarded as a means of augmenting the learning experience [2]. Studies have shown that educational VR can enhance experiential learning, facilitate the demonstration of complex concepts, and increase student engagement and motivation [4, 15]. Furthermore, research has indicated that VR can generate more profound experience memory [14]. VR technology offers the ability to create immersive experiences that are difficult to replicate through traditional e-learning and blended learning methods, allowing learners to more effectively engage with objects and phenomena with less cognitive effort [13].

Although high-immersive environments have gained increased attention [11], empirical studies and meta-analyses have demonstrated that while high-immersive VR environments may elicit a greater sense of presence, they do not necessarily result in enhanced learning outcomes [7]. It has been suggested that the use of low-immersive or desktop VR environments can lead to improved cognitive outcomes and positive learning attitudes [8, 10]. Additionally, the high-immersive VR environments may pose challenges for some learners, particularly those engaging in distance learning, due to equipment limitations. Desktop VR, which only requires the use of a monitor, keyboard, and mouse, is a more accessible and cost-effective option [19].

Interactive learning has been extensively studied, such as human interaction in online environments and higher learning perception [12], higher curriculums satisfaction among students and teachers [5, 17], and the improvement of students' performance [6]. 3D multi-user virtual environments (MUVEs) offer a shared, realistic, and immersive space in which learners can explore, interact, and modify the virtual world through the use of avatars. Furthermore, the use of text, voice, or animated interactions in MUVEs enables learners to engage in collaborative activities [9]. Users move freely in the 3D MUVEs environment using avatars and can see others, which

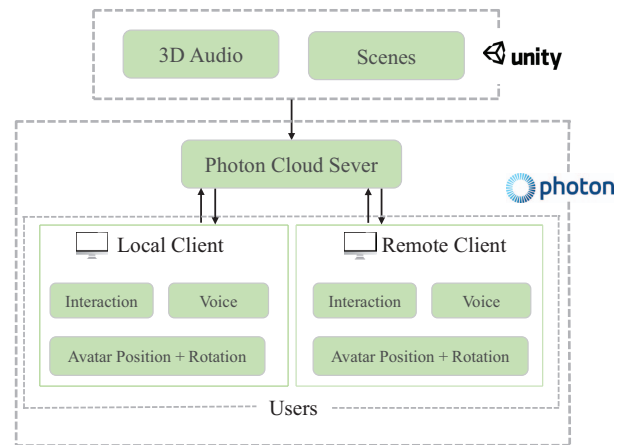


Figure 1: Multi-user learning system structure diagram.

can increase the sense of immersion and social presence [3, 20]. The sense of presence and social interaction provided by 3D MUVEs connect distributed team members, enhancing collaboration and innovation within virtual teams [1].

In this study, we propose a desktop VR interactive learning system based on a 3D MUVE for the study of butterflies, allowing multiple users to participate in a shared virtual environment and interact with each other through avatars and voice communication. A user study was conducted to evaluate the system's effectiveness, receiving positive ratings in terms of perception, system effectiveness, learning experience, enjoyment and attractiveness. However, areas for improvement were identified in terms of realism and object manipulation capabilities. The findings have implications for the design of educational multi-user VR environments.

2 DESIGN AND IMPLEMENTATION OF 3D MUVES SYSTEM

This study proposes a desktop VR interactive learning system based on a 3D MUVE, for facilitating collaborative learning and knowledge building among students. As shown in Fig. 1, the system is built by using Unity¹, a cross-platform game engine with a built-in IDE, and is published to the Windows platform. The system uses a Photon cloud server² to enable online functionality, allowing multiple users to access the system via the network. The system includes the incorporation of avatars, allowing users to explore and learn from a first-person perspective. Additionally, the position and rotation information of the avatars are transmitted through the server to achieve online synchronization. The system also supports real-time voice communication among users.

2.1 Networking Setup

The system was built upon PUN2 (Photon Unity Networking), a Unity built-in asset package for online multiplayer games, that is

¹<https://unity.com/developer-tools>

²<https://www.photonengine.com/>

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linked to the photon cloud. It offers players rooms and lobbies. Connecting to a lobby is equivalent to connecting to a session server. Connecting to a room is equivalent to connecting to a game server. As a result, a user must first connect to the lobby before proceeding to the room. The free version of PUN v2 is used in our system.

2.2 Scenes and Avatars

The present system features four distinct scenes: the login scene, the lobby scene, the outdoor scene, and the school scene. The models used within these scenes were created using 3ds max or sourced from the Unity resource store.

The login scene comprises a login interface, where users have the option to "CONNECT WITH NAME" or "CONNECT ANONYMOUSLY". Additionally, there are operational instructions provided beneath the login interface. Upon selecting "CONNECT ANONYMOUSLY", the user is directed to the lobby scene. If the user selects "CONNECT WITH NAME", they are prompted to enter a custom name before being directed to the lobby scene.

Upon entering the lobby scene, the user is presented with the default avatar of the system. By pressing the "M" key, a menu interface becomes accessible. This interface offers the options of "WORLDS" or "CHANGE AVATAR". Selecting "CHANGE AVATAR" displays an avatar platform, from which the user can choose from five available avatars. Choosing "WORLDS" opens the world user interface, providing the user the option to enter either the outdoor scene or the school scene.

The outdoor scene is designed for the purpose of exploration and learning. Various types of butterflies are strategically placed within the scene. By clicking on a butterfly, an introduction panel providing information about the specific butterfly is activated (see Fig. 2(a)). This scene encourages collaborative learning among users.

The school scene features a "Today's Discussion" panel and various activities (see Fig. 2(b)). Users are able to engage in collective activities and discussions within this scene.

2.3 Multi-User Interaction

The integration of avatars within single-user VR environments is not observed to have a significant impact on the level of immersion experienced by users. However, this may change within the context of MUEs. In such systems, users are not only required to track their own avatars, but also those of other users. To achieve this, the position and rotation of the user's head, hands, and body are transmitted over the network, and each client updates the position and rotation of the corresponding avatar in real time. The PhotonView and PhotonTransformView components are employed to record this information.

The interaction between users and scene objects within a MUE is more complex than that of single-user VR, and updating the information only on the local client is inadequate. In consideration of system performance, access to objects is restricted by granting permission to a single client at a time. Physical information is only tracked on clients that have been granted permission to interact with an object and is subsequently transmitted to other clients. This approach is implemented to optimize system performance.

2.4 Multi-User Voice Chat

The Photon voice component suite consists primarily of the Photon Voice Speaker and the Photon Voice Recorder. The Photon Voice Speaker is responsible for the replay and playback of remotely received audio, while the Photon Voice Recorder records and transmits local audio. Upon joining a room, users will automatically establish a connection with the Photon Voice system.

3 USER STUDY

We evaluated the MUE system with a sample of 13 university students (8 male and 5 female) who had no prior experience with



(a)



(b)

Figure 2: Learning Tasks: (a) Learning materials; (b) Discussion materials.

interactive learning in a MUE. The study was conducted on the participants' personal computers, through the use of a chat group for data distribution and collection.

Before the experiment, we sent the electronic version of the consent form to the chat group, and all participants agreed to participate in the experiment. The participants were instructed to download the application and prepare their earphones. An explanation of the experimental procedure was provided. The participants filled out a six-question pre-test questionnaire about butterflies. The tester then provided a demonstration of the program to familiarize the participants with its operation.

During the experiment, the participants used the MUE and user manual to explore and learn. They first entered an outdoor scene to learn about butterflies, and then entered a school scene to discuss. Finally, they completed the post-test and a user experience questionnaire after learning in MUE.

4 RESULTS

4.1 Learning Performance

The total scores for the pre-test and post-test assessing knowledge of butterflies were 6 points. The average score for the 13 participants in the pre-test was 2.46, whereas in the post-test the average score was 4.46. This indicates an improvement in the students' knowledge after they learn with MUE.

The normality of the pre-test, post-test and the difference between the two tests was evaluated. Given the sample size of 13, the Shapiro-Wilk test was used to analyze the pre-test ($p = 0.025$), post-test ($p = 0.048$) and their difference ($p = 0.455$), as well as the histogram. The results suggest that the data is close to a normal distribution. A paired sample t-test was then conducted on the sample data, the results of which are presented in Table 1. There was a statistically significant difference between the pre-test and post-test ($p < 0.01$), indicating that the system effectively promotes learning.

Table 1: T-test Results of pre-test and post-test.

Test	Mean $\pm$ SD	T value	Sig.(two-tailed)
Pre-test	2.46 $\pm$ 0.87	-4.899	0.000
Post-test	4.462 $\pm$ 1.33		

4.2 User Experience

The user experience questionnaire used in this study comprised six dimensions: perception of the quality of VR technology, perception of the quality of VR use, system effectiveness, social experience, learning experience, enjoyment and attractiveness. The scale design for the user experience questionnaire was based on the User Experience Questionnaire Handbook [16] which is widely used as a reference for the design of user experience scales. The design of the VR experience dimension was informed by Tarng's scale [18] on virtual laboratory research. The questionnaire consisted of 21 questions in total, which can be found in Table 2. Participants were asked to respond to Likert items on a five-point scale, with 1 indicating a negative response and 5 indicating a positive response, or semantic differential on a five-point.

The results of the internal consistency analysis of the questionnaire indicate a satisfactory level of reliability, as evidenced by a Cronbach's alpha coefficient of 0.86. As illustrated in Fig. 3, the questionnaire consisted of 21 questions and 6 dimensions. The scores for the dimensions of VR technology quality perception and VR use quality perception were found to be relatively lower, particularly in regard to the manipulation of objects within the VR environment. However, the dimensions of system effectiveness, social experience, and learning experience yielded higher scores, although the score for the dimension of system learning knowledge was comparatively lower, suggesting a need for improvement in the system's learning activities. Notably, the dimensions of enjoyment and attractiveness received high scores, indicating that the learning system effectively engages learners and improves attention.

- **VR technology quality perception:** questions 1, 2, 3 and 4. The median score of question 1 is 3 and the median scores (questions 2, 3 and 4) are all above 3.
- **VR use quality perception:** questions 5, 6, 7, 8 and 9. The median scores (questions 5, 6, 8 and 9) are all above 3 and the score of question 7 is 3.
- **system effectiveness:** questions 10, 11, and 12. The median scores are all 4.
- **social experience:** questions 13, 14 and 15. The median score of question 13 is 5 and the median scores (questions 14 and 15) are all 4.
- **learning experience:** questions 16, 17 and 18. The median scores of question (questions 16 and 18) are all 4 and the median score of question 17 is 5.
- **enjoyment and attractiveness:** questions 19, 20 and 21. The median scores of question (questions 19 and 21) are all 5 and the median score of question 20 is 4.

5 DISCUSSION

The results of the pre-test and post-test analysis indicate a significant improvement in scores, suggesting that the virtual reality interactive learning system effectively promotes student learning. User experience evaluations revealed the good performance of the learning system in four dimensions: system effectiveness, social experience, learning experience, enjoyment, and attractiveness. However,

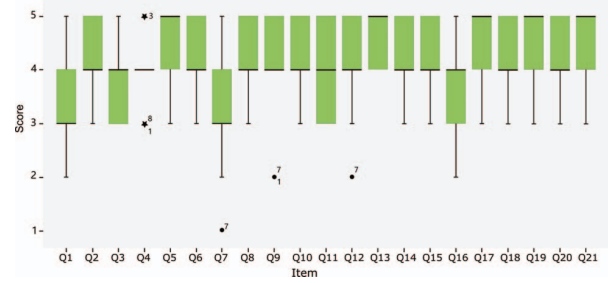


Figure 3: User Experience Result

it is important to note that there is room for improvement in the dimensions of VR technology quality perception and use quality experience. Specifically, the realism of the system and the experience of manipulating objects within the VR environment require further enhancement. Open-ended feedback from learners suggests that the system should incorporate additional interactive features and enhance the realism of the VR experience. Additionally, learners recommend incorporating more diverse and interactive learning content, such as incorporating game-based elements related to the learning content to aid in the mastery of knowledge.

6 CONCLUSION AND FUTURE WORK

In this study, a desktop VR interactive learning system based on a 3D MUVE was designed and developed. Through a user study, the effectiveness of the system in promoting learning and the user experience was evaluated. The results indicated that by designing appropriate learning activities and social interactions, the social and learning needs of students can be met. The system was found to be well-received by users, and the novel and interesting form of the system was found to be attractive. The research results suggest that the desktop VR interactive learning system based on a 3D MUVE has the potential to reduce social distance among learners and improve the learning experience, offering new opportunities for online education.

However, the system also presented certain limitations. The interaction between users and the virtual environment was found to be limited, the immersion of the system needs to be improved, the ability to manipulate objects within the virtual environment needs to be enhanced, and the design of social activities and learning content should be further developed.

In future work, we plan to investigate the use of immersive VR technology, specifically multi-user VR, as a replacement for the current desktop VR system. The goal of this transition is to enhance the realism and user experience while maintaining learning effectiveness. Furthermore, we plan to continue to use this system as a platform for the development of high-quality user interactions and exploration activities that provide learners with increased autonomy and a more authentic and free social experience. Additionally, systematic evaluations and controlled experiments will be conducted to explore the underlying design principles pertaining to pedagogy.

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Table 2: The user experience questionnaire.

dimensions	Statements
VR technology quality perception	Q1: The degree of realism I experienced in a VR environment. Q2: The degree of enjoyment I experienced in a VR environment. Q3: I feel the quality of the VR environment. Q4: I feel the quality of the VR technology.
VR use quality perception	Q5: It is not difficult for me to operate the menu in VR. Q6: It is easy for physical movement in VR. Q7: It is easy to manipulate objects in VR. Q8: It is smooth for the scene transitions in VR. Q9: It is easy to operate mouse/keyboard for interaction in VR.
System effectiveness	Q10: I think the system is valuable. Q11: I would like to recommend this system to others. Q12: I think this system is practical.
Social experience	Q13: This system supports social activities between learners. Q14: I feel comfortable interacting with other learners. Q15: I enjoy using this system with other learners.
Learning experience	Q16: I can learn knowledge easily in the system. Q17: I am interested in exploring the learning content in the system. Q18: The communication function of the system can help me solve study problems.
Enjoyment and attractiveness	Q19: I enjoy using this system. Q20: I think the system is attractive. Q21: I think this system is novel.

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